Being the lowest energy state, |0> is the ground state, and its energy is $E_0 = \omega\hbar/2$. It is an eigenvector of **N** with an eigenvalue 0. We often say that the ground state is *annihilated* by $\mathbf{a}^-$.

So you see, the abstract construction of $\mathbf{a}^+$, $\mathbf{a}^-$, and **N** paid off. It allowed us to find the entire spectrum of harmonic oscillator energy levels without solving a single difficult equation. This spectrum consists of the energy values,

$$\begin{aligned} E_n &= \omega\hbar(n + 1/2) \\ &= \omega\hbar\ (\ 1/2,\ 3/2,\ 5/2, \ldots\ ). \qquad (10.30) \end{aligned}$$

This quantization of harmonic oscillator energy levels was one of the first results of quantum mechanics, and arguably the most important. The hydrogen atom is a wonderful example of quantum mechanics, but it is, after all, just the hydrogen atom. The harmonic oscillator, on the other hand, shows up everywhere, from crystal vibrations to electric circuits to electromagnetic waves. The list goes on. Even macroscopic oscillators, like a child on a swing, have quantized energy levels, but the presence of Planck's constant in Eq. 10.30 means that the spacing between levels is so tiny that they are completely undetectable.

The unending spectrum of positive energy levels for a harmonic oscillator is sometimes called a tower, and sometimes called a ladder. It is illustrated schematically in Fig. 10.2.

10.7 BACK TO WAVE FUNCTIONS

This exercise has amply demonstrated the remarkable power of operator algebras, and the operator method is indeed remarkable. But it's also very abstract. Is it useful in helping us find wave functions, which are more concrete and easier to visualize? Absolutely.

Let's begin with the ground state. We just saw in Eq. 10.29 that the ground state is the unique state that is annihilated by $\mathbf{a}^-$. Now, let's rewrite